Week 7 complete solutions

Problem-sheet solutions

Each solution states the intended method, relevant conditions, complete reasoning, mark allocation and common mistakes. Award method credit after a purely arithmetic slip when the setup remains valid.

W07-Q01-PA (M10)

Method and conditions. Use the stated equality-constraint encoding and solve stationarity together with feasibility. The logarithm domain requires $x > 0$ and $y > 0$.

The Lagrangian is

$$\mathcal{L}(x, y, \mu) = \log x + 2 \log y + \mu(12 - x - 2y).$$

The first-order and feasibility conditions are

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{x} - \mu = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{2}{y} - 2\mu = 0,$$

and

$$12 - x - 2y = 0.$$

The first two equations give

$$\mu = \frac{1}{x} = \frac{1}{y},$$

so $x = y$. Feasibility then gives $3x = 12$.

Final result.

$$(x^*, y^*) = (4, 4), \qquad \mu^* = \frac{1}{4}.$$

Both variables satisfy the positive-domain restrictions.

Marks (5). Correct Lagrangian, 1; two correct stationarity conditions, 1; feasibility and equality of the two variables, 1; correct candidate, 1; correct multiplier and domain check, 1.

Common mistakes. Omitting the factor two in the constraint derivative with respect to y; using a minus multiplier term despite the stated encoding; failing to check the logarithm domain.

W07-Q01-PB (M10)

Method and conditions. Establish sufficiency from strict concavity and the affine constraint. Solve the parameterised problem, form the value function and compare its derivative with the partial derivative of the Lagrangian.

The Hessian of the objective on the positive orthant is

$$\begin{bmatrix} -1/x^2 & 0 \\ 0 & -2/y^2 \end{bmatrix},$$

which is negative definite. The objective is therefore strictly concave. The feasible set defined by the affine equality and positivity is convex, so the feasible first-order candidate is the unique global maximiser.

With $a > 0$ in place of 12, the same first-order equations give $x = y$, and the constraint gives

$$x^*(a) = y^*(a) = \frac{a}{3}, \qquad \mu^*(a) = \frac{3}{a}.$$

The value function is

$$\begin{aligned} V(a) &= \log\left(\frac{a}{3}\right) + 2\log\left(\frac{a}{3}\right) \\ &= 3\log\left(\frac{a}{3}\right). \end{aligned}$$

Hence

$$V'(a) = \frac{3}{a}.$$

Because the parameter enters the constraint as $a - x - 2y = 0$,

$$\left.\frac{\partial \mathcal{L}}{\partial a}\right|_* = \mu^*(a) = \frac{3}{a}.$$

At $a = 12$, both the envelope derivative and the multiplier equal $1/4$.

Final result and interpretation. The solution in Part A is the unique global maximiser. Under the stated encoding, a small increase $\mathrm{d}a > 0$ increases the optimised objective by approximately $(1/4)\mathrm{d}a$ at $a = 12$.

Marks (7). Strict-concavity and affine-feasible-set sufficiency, 2; parameterised optimiser and multiplier, 2; correct value function, 1; envelope agreement at $a = 12$, 1; sign interpretation tied to the encoding, 1.

Common mistakes. Treating first-order conditions alone as sufficient; differentiating the unoptimised objective while holding the choice variables arbitrarily fixed; interpreting the multiplier without naming the constraint encoding.

`W07-Q02-PA (M10)`

Method and conditions. Form the Lagrangian using $h(x, y, a) = a - x - y$, solve the first-order system as a function of a, and use strict concavity for sufficiency.

$$\mathcal{L}(x, y, \mu; a) = -(x-2)^2 - (y-1)^2 + \mu(a - x - y).$$

Stationarity and feasibility give

$$-2(x-2) - \mu = 0, \qquad -2(y-1) - \mu = 0, \qquad a - x - y = 0.$$

Thus

$$x = 2 - \frac{\mu}{2}, \qquad y = 1 - \frac{\mu}{2}.$$

Substitution into feasibility yields $3 - \mu = a$, so

$$\mu^*(a) = 3 - a.$$

Therefore

$$x^*(a) = \frac{a+1}{2}, \qquad y^*(a) = \frac{a-1}{2}.$$

Both squared deviations equal $(a-3)^2/4$, so

$$V(a) = -\frac{(a-3)^2}{2}.$$

The objective Hessian is $-2\mathbf{I}_2$, which is negative definite, and the equality constraint is affine. The candidate is therefore the unique global maximiser for every $a \in \mathbb{R}$.

Final result.

$$x^*(a) = \frac{a+1}{2}, \quad y^*(a) = \frac{a-1}{2}, \quad \mu^*(a) = 3 - a, \quad V(a) = -\frac{(a-3)^2}{2}.$$

Marks (7). Correct Lagrangian and first-order system, 2; correct multiplier, 1; two correct optimisers, 2; correct value function, 1; sufficiency argument, 1.

Common mistakes. Losing the minus sign when differentiating a squared loss; imposing an unstated non-negativity restriction on x or y; claiming a global maximum without using strict concavity.

`W07-Q02-PB (M10)`

Method and conditions. Apply the envelope theorem under the stated constraint encoding, then verify the derivative directly.

Because a enters only through $\mu(a - x - y)$,

$$V'(a) = \left.\frac{\partial \mathcal{L}}{\partial a}\right|_{*} = \mu^*(a) = 3 - a.$$

Direct differentiation gives the same result:

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[-\frac{(a-3)^2}{2}\right] = -(a-3) = 3 - a.$$

If the same constraint is written $x + y - a = 0$, the Lagrangian term is $\tilde{\mu}(x + y - a)$. The first-order conditions then give

$$\tilde{\mu}^*(a) = -\mu^*(a) = a - 3.$$

The envelope derivative is nevertheless unchanged because

$$\frac{\partial \mathcal{L}}{\partial a} = -\tilde{\mu}^*(a) = 3 - a.$$

Final result and interpretation. The value derivative is $3 - a$. The multiplier changes sign when the equality constraint is multiplied by -1, while the value function and its economic interpretation do not.

Marks (4). Correct envelope derivative, 1; direct reconciliation, 1; correct multiplier sign under the reversed encoding, 1; correct explanation that the value derivative is invariant, 1.

Common mistakes. Keeping the same multiplier after reversing the constraint; changing the value derivative when only the encoding changes; using a total derivative where the envelope theorem requires a partial derivative of the Lagrangian.

`W07-Q03-PA (M10)`

Method and conditions. Use one multiplier for each equality, verify linear independence of the equality-constraint gradients, solve the stationarity system and use strict concavity for uniqueness.

The Lagrangian is

$$\mathcal{L}(x, y, z, \mu, \nu; a) = -(x^2 + y^2 + z^2) + \mu(a - x - y - z) + \nu(x - y).$$

Stationarity gives

$$-2x - \mu + \nu = 0,$$

$$-2y - \mu - \nu = 0,$$

$$-2z - \mu = 0.$$

Feasibility requires

$$x + y + z = a, \qquad x - y = 0.$$

The second equality gives $x = y$. Subtracting the first two stationarity conditions then gives $2\nu = 0$, so $\nu = 0$. The first and third stationarity conditions now imply $x = z$. Hence

$$x^*(a) = y^*(a) = z^*(a) = \frac{a}{3}.$$

The third stationarity condition gives

$$\mu^*(a) = -2z^*(a) = -\frac{2a}{3}, \qquad \nu^*(a) = 0.$$

The equality-constraint gradients are

$$\nabla(a - x - y - z) = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}, \qquad \nabla(x - y) = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

They are linearly independent, so the regularity condition holds. The objective Hessian is $-2\mathbf{I}_3$, which is negative definite, and the constraints are affine. Thus the candidate is the unique global maximiser.

The value function is

$$V(a) = -3\left(\frac{a}{3}\right)^2 = -\frac{a^2}{3}.$$

Therefore

$$V'(a) = -\frac{2a}{3} = \mu^*(a) = \left.\frac{\partial \mathcal{L}}{\partial a}\right|_*.$$

At $a = 6$,

$$(x^*, y^*, z^*) = (2, 2, 2), \qquad \mu^* = -4, \qquad \nu^* = 0, \qquad V(6) = -12, \qquad V'(6) = -4.$$

Marks (10). Correct Lagrangian and five conditions, 2; correct optimisers, 2; correct two multipliers, 1; regularity check, 1; strict concavity and uniqueness, 1; value function, 1; envelope verification, 1; complete evaluation at $a = 6$, 1.

Common mistakes. Using one multiplier for two independent equalities; omitting the equality-gradient regularity check; assuming the first multiplier must be positive; failing to connect the envelope derivative to the specific constraint containing a.

`W07-Q04-PA (M11)`

Method and conditions. Write both bounds as explicit non-negative constraint functions and apply the course maximisation convention.

Define

$$g_1(x) = x, \qquad g_2(x) = 3 - x.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda_1, \lambda_2) = 8x - x^2 + \lambda_1 x + \lambda_2(3 - x).$$

The complete Kuhn–Tucker conditions are:

$$8 - 2x + \lambda_1 - \lambda_2 = 0 \quad \text{(stationarity)},$$

$$x \geq 0, \qquad 3 - x \geq 0 \quad \text{(primal feasibility)},$$

$$\lambda_1 \geq 0, \qquad \lambda_2 \geq 0 \quad \text{(dual feasibility)},$$

and

$$\lambda_1 x = 0, \qquad \lambda_2(3 - x) = 0 \quad \text{(complementary slackness)}.$$

Marks (5). Two correctly encoded constraints, 1; correct Lagrangian, 1; stationarity, 1; primal and dual feasibility, 1; both complementary-slackness equations, 1.

Common mistakes. Writing the upper bound as $x - 3 \geq 0$; using a minus sign before a multiplier term; leaving $x \geq 0$ as an unstated domain restriction.

`W07-Q04-PB (M11)`

Method and conditions. Examine the interior and each endpoint, reject infeasible or dual-infeasible branches, and then apply a sufficiency condition.

If both constraints are slack, complementary slackness gives $\lambda_1 = \lambda_2 = 0$. Stationarity then gives $x = 4$, which violates $3 - x \geq 0$.

At the lower endpoint $x = 0$, the upper bound is slack, so $\lambda_2 = 0$. Stationarity would require

$$8 + \lambda_1 = 0,$$

or $\lambda_1 = -8$, violating dual feasibility.

At the upper endpoint $x = 3$, the lower bound is slack, so $\lambda_1 = 0$. Stationarity gives

$$8 - 2(3) - \lambda_2 = 0,$$

so $\lambda_2 = 2$, which is dual feasible. Both endpoints cannot bind simultaneously.

Final result.

$$x^* = 3, \qquad \lambda_1^* = 0, \qquad \lambda_2^* = 2.$$

The objective has second derivative $-2 < 0$, so it is strictly concave, and the feasible interval is convex. The feasible Kuhn–Tucker point is therefore the unique global maximiser.

Marks (7). Interior branch and infeasibility, 1; lower-bound branch and dual-feasibility rejection, 2; upper-bound solution, 2; complete multiplier report, 1; sufficiency and uniqueness, 1.

Common mistakes. Accepting the infeasible unconstrained candidate; setting the multiplier on a binding constraint to zero automatically; failing to reject a negative multiplier; omitting the global sufficiency argument.

`W07-Q05-PA (M11)`

Method and conditions. Encode all three inequalities in $g_j(x, y) \geq 0$ form. On the requested branch, use complementary slackness before solving stationarity and feasibility.

Define

$$g_1(x, y) = x, \qquad g_2(x, y) = y, \qquad g_3(x, y) = 4 - x - y.$$

The Lagrangian is

$$\mathcal{L} = 6x + 4y - x^2 - y^2 + \lambda_1 x + \lambda_2 y + \lambda_3(4 - x - y).$$

Stationarity gives

$$6 - 2x + \lambda_1 - \lambda_3 = 0,$$

$$4 - 2y + \lambda_2 - \lambda_3 = 0.$$

Primal and dual feasibility are

$$x \geq 0, \qquad y \geq 0, \qquad 4 - x - y \geq 0,$$

$$\lambda_1, \lambda_2, \lambda_3 \geq 0.$$

Complementary slackness is

$$\lambda_1 x = 0, \qquad \lambda_2 y = 0, \qquad \lambda_3(4 - x - y) = 0.$$

On the branch $x > 0$, $y > 0$ and $x + y = 4$, complementary slackness gives $\lambda_1 = \lambda_2 = 0$. Stationarity becomes

$$6 - 2x = \lambda_3, \qquad 4 - 2y = \lambda_3.$$

Thus $x - y = 1$. Combining this with $x + y = 4$ gives

$$x = \frac{5}{2}, \qquad y = \frac{3}{2}.$$

Either stationarity equation then gives $\lambda_3 = 1$.

Final result for this branch.

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = \left(\frac{5}{2}, \frac{3}{2}, 0, 0, 1\right).$$

It satisfies primal feasibility, dual feasibility and complementary slackness.

Marks (7). Three correctly encoded constraints and Lagrangian, 2; stationarity, 1; primal, dual and complementary conditions, 1; correct active-set reduction, 1; correct x, y, 1; correct multipliers and feasibility check, 1.

Common mistakes. Omitting multiplier terms for variable non-negativity; reversing the sign of the resource constraint; setting the resource multiplier to zero even though the resource binds; solving the stationarity equations without imposing $x + y = 4$.

`W07-Q05-PB (M11)`

Method and conditions. Check every qualitatively different active-set branch, using a negative required multiplier or primal infeasibility to reject a branch.

If the resource is slack and both inputs are positive, all multipliers are zero. Stationarity gives the unconstrained candidate

$$(x, y) = (3, 2),$$

which violates $x + y \leq 4$.

Suppose $x = 0$. If the resource is slack and $y > 0$, then $\lambda_2 = \lambda_3 = 0$, and the y-stationarity equation gives $y = 2$. The x-stationarity equation then requires $\lambda_1 = -6$, violating dual feasibility. If the resource binds, then $y = 4$ and $\lambda_2 = 0$. The y-stationarity equation requires $\lambda_3 = -4$, again violating dual feasibility. At the origin, the resource is slack and x-stationarity likewise requires a negative λ_1.

Suppose $y = 0$. If the resource is slack and $x > 0$, stationarity gives $x = 3$, but the y-stationarity equation requires $\lambda_2 = -4$. If the resource binds, then $x = 4$ and $\lambda_1 = 0$; the x-stationarity equation requires $\lambda_3 = -2$. Both branches violate dual feasibility.

Therefore the point found in Part A is the only Kuhn–Tucker point.

The objective Hessian is

$$\mathbf{H}_f = -2\mathbf{I}_2,$$

which is negative definite. The objective is strictly concave, and the feasible set is the convex intersection of three affine half-spaces.

Final result. The unique global maximiser is

$$(x^*, y^*) = \left(\frac{5}{2}, \frac{3}{2}\right),$$

with multipliers $(0, 0, 1)$.

Marks (6). Rejection of the slack-resource branch, 1; correct $x = 0$ branch checks, 1; correct $y = 0$ branch checks, 1; conclusion that no other Kuhn–Tucker point remains, 1; strict-concavity and convex-feasible-set sufficiency, 1; unique global conclusion, 1.

Common mistakes. Checking objective corners without checking multiplier signs; overlooking the resource-slack branch; treating a negative multiplier as admissible; claiming uniqueness from stationarity alone.

`W07-Q06-PA (M11)`

Method and conditions. Correct the feasible-set representation and the Lagrangian sign before applying the Kuhn–Tucker conditions.

The student's work has three connected faults:

- $x - 1 \geq 0$ represents $x \geq 1$, not $x \leq 1$;
- the lower bound $x \geq 0$ has been omitted; and
- the course convention uses a plus multiplier term for a constraint written $g_j(x) \geq 0$.

Use

$$g_1(x) = x, \qquad g_2(x) = 1 - x.$$

Then

$$\mathcal{L}(x, \lambda_1, \lambda_2) = 4x - x^2 + \lambda_1 x + \lambda_2(1 - x).$$

The complete conditions are

$$4 - 2x + \lambda_1 - \lambda_2 = 0,$$

$$x \geq 0, \qquad 1 - x \geq 0, \qquad \lambda_1, \lambda_2 \geq 0,$$

$$\lambda_1 x = 0, \qquad \lambda_2(1 - x) = 0.$$

The interior branch gives $x = 2$, which is infeasible. At $x = 0$, stationarity would require $\lambda_1 = -4$, which is dual infeasible. At $x = 1$, complementary slackness gives $\lambda_1 = 0$, and stationarity gives $\lambda_2 = 2$.

Final result.

$$x^* = 1, \qquad \lambda_1^* = 0, \qquad \lambda_2^* = 2.$$

Because the objective is strictly concave on the convex interval $[0, 1]$, this is the unique global maximiser.

Marks (6). Identification of the incorrect upper-bound encoding and omitted lower bound, 1; identification of the Lagrangian-sign error, 1; correct constraints and Lagrangian, 1; complete Kuhn–Tucker conditions, 1; correct point and multipliers, 1; sufficiency conclusion, 1.

Common mistakes. Treating $x - 1 \geq 0$ as an upper bound; omitting the lower-bound multiplier; retaining a minus multiplier term after converting to $g_j \geq 0$; accepting a negative multiplier at $x = 0$.

`W07-Q07-PA (S7)`

Method and conditions. Separate the random estimator from its realised estimate. Use linearity of expectation, the iid sample-mean variance and the bias–variance decomposition of mean squared error.

The parameter and estimand are both the population mean θ. Before observation,

$$\hat{\theta}_n = \bar{X} + \frac{1}{n}$$

is the estimator. For the observed data,

$$\bar{x} = \frac{1 + 2 + 3 + 4}{4} = \frac{5}{2},$$

so the observed estimate is

$$\hat{\theta}_4(\mathbf{x}) = \frac{5}{2} + \frac{1}{4} = \frac{11}{4}.$$

The estimator's expectation is

$$\mathbb{E}_\theta[\hat{\theta}_n] = \mathbb{E}_\theta[\bar{X}] + \frac{1}{n} = \theta + \frac{1}{n}.$$

Therefore

$$\mathrm{Bias}_\theta(\hat{\theta}_n) = \frac{1}{n}.$$

The added term is non-random, so

$$\mathrm{Var}_\theta(\hat{\theta}_n) = \mathrm{Var}_\theta(\bar{X}) = \frac{\sigma^2}{n}.$$

Thus

$$\mathrm{MSE}_\theta(\hat{\theta}_n) = \frac{\sigma^2}{n} + \frac{1}{n^2}.$$

This expression tends to zero. For every $\varepsilon > 0$, Markov's inequality applied to the squared error gives

$$\mathbb{P}_\theta\left(|\hat{\theta}_n - \theta| > \varepsilon\right) \leq \frac{\mathrm{MSE}_\theta(\hat{\theta}_n)}{\varepsilon^2} \longrightarrow 0.$$

Hence $\hat{\theta}_n$ is consistent for θ.

Final result. The observed estimate is $11/4$; the bias is $1/n$, the variance is σ^2/n, the mean squared error is $\sigma^2/n + 1/n^2$, and the estimator is consistent.

Marks (6). Four-way terminology distinction and observed estimate, 2; bias, 1; variance, 1; mean squared error, 1; valid consistency proof, 1.

Common mistakes. Calling the observed number an estimator; adding $1/n$ to the variance; omitting squared bias from mean squared error; calling vanishing bias alone a proof of consistency.

`W07-Q08-PA (S7)`

Method and conditions. Equate the population first moment to the sample first moment, then use expectation and the law of large numbers to study the resulting estimator.

The method-of-moments equation is

$$\frac{\hat{\theta}_{\mathrm{MOM}}}{2} = \bar{X},$$

so

$$\hat{\theta}_{\mathrm{MOM}} = 2\bar{X}.$$

For the observed sample,

$$\bar{x} = \frac{1+2+2+3}{4} = 2,$$

and the estimate is

$$\hat{\theta}_{\mathrm{MOM}}(\mathbf{x}) = 4.$$

The fitted support is $[0, 4]$, which contains every observation, including the sample maximum 3.

The estimator is unbiased because

$$\mathbb{E}_\theta[2\bar{X}] = 2\frac{\theta}{2} = \theta.$$

The law of large numbers gives

$$\bar{X} \xrightarrow{p} \frac{\theta}{2}.$$

Multiplying by two gives

$$2\bar{X} \xrightarrow{p} \theta,$$

so the estimator is consistent.

Final result.

$$\hat{\theta}_{\mathrm{MOM}} = 2\bar{X},$$

with observed estimate 4; it is unbiased and consistent.

Marks (6). Correct moment equation, 1; correct estimator, 1; numerical estimate, 1; fitted-support check, 1; bias derivation, 1; consistency argument, 1.

Common mistakes. Equating θ rather than $\theta/2$ to $\bar{X}$; reporting $\bar{X}$ as the estimator; failing to compare the fitted upper endpoint with the sample maximum; asserting consistency from unbiasedness alone.

`W07-Q09-PA (S7)`

Method and conditions. Form the likelihood on the closed parameter space $[0, 1]$. Use the log-likelihood for an interior sample and inspect the likelihood directly for boundary samples.

Independence gives

$$L(p; \mathbf{x}) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^s(1-p)^{n-s}, \qquad p \in [0, 1].$$

If $0 < s < n$, the log-likelihood is

$$\ell(p; \mathbf{x}) = s \log p + (n-s) \log(1-p), \qquad 0 < p < 1.$$

Its derivative is

$$\frac{\partial \ell}{\partial p} = \frac{s}{p} - \frac{n-s}{1-p}.$$

Setting this equal to zero gives

$$s(1-p) = p(n-s),$$

and hence $p = s/n$. Moreover,

$$\frac{\partial^2 \ell}{\partial p^2} = -\frac{s}{p^2} - \frac{n-s}{(1-p)^2} < 0,$$

so the interior candidate is the unique maximum.

If $s = 0$, then

$$L(p; \mathbf{x}) = (1-p)^n,$$

which is maximised at $p = 0$. If $s = n$, then

$$L(p; \mathbf{x}) = p^n,$$

which is maximised at $p = 1$.

Final result. In all cases,

$$\hat{p}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} X_i = \bar{X}.$$

The formula includes the endpoint estimates zero and one.

Marks (8). Correct likelihood and parameter space, 2; interior log-likelihood and score, 2; interior estimator, 1; curvature maximum check, 1; correct $s = 0$ boundary, 1; correct $s = n$ boundary and unified result, 1.

Common mistakes. Omitting the parameter-space endpoints; differentiating $\log p$ at $p = 0$; reporting the score equation without a maximum check; ignoring the all-zero or all-one samples.

`W07-Q09-PB (S7)`

Method and conditions. Form the joint density for an observed sample in the support, maximise over $\lambda > 0$, and compare with the first-moment equation.

For a valid observed sample, every $x_i \geq 0$. If any observation is negative, the sample is outside the stated model support and the estimation problem must be reconsidered. For $x_i \geq 0$, write $T = \sum_{i=1}^{n} x_i$. Independence gives

$$L(\lambda; \mathbf{x}) = \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} x_i\right), \qquad \lambda > 0.$$

The log-likelihood is

$$\ell(\lambda; \mathbf{x}) = n \log \lambda - \lambda \sum_{i=1}^{n} x_i.$$

For $T > 0$, the score equation is

$$\frac{\partial \ell}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0,$$

so

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} X_i} = \frac{1}{\bar{X}}.$$

The second derivative is

$$\frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{n}{\lambda^2} < 0.$$

Also, the log-likelihood tends to $-\infty$ as $\lambda \downarrow 0$ and, because $T > 0$, as $\lambda \to \infty$. Thus the stationary point is the unique global maximum when $T > 0$.

The remaining valid-support case is $T = 0$, which means every observed value is zero. Then

$$L(\lambda; \mathbf{x}) = \lambda^n.$$

This likelihood is unbounded as $\lambda \to \infty$, so no finite MLE exists on $\lambda > 0$. This case has probability zero under the continuous exponential model, but it still has to be handled as a possible realised data vector.

For the observed sample, $n = 4$ and

$$\sum_{i=1}^{4} x_i = \frac{1}{2} + 1 + \frac{3}{2} + 1 = 4.$$

Therefore

$$\hat{\lambda}_{\mathrm{MLE}}(\mathbf{x}) = 1.$$

The population mean is $1/\lambda$, so the method-of-moments equation is

$$\frac{1}{\hat{\lambda}_{\mathrm{MOM}}} = \bar{X}.$$

When $T > 0$, it gives

$$\hat{\lambda}_{\mathrm{MOM}} = \frac{1}{\bar{X}} = \frac{n}{\sum_i X_i}.$$

When $T = 0$, the moment equation $1/\lambda = 0$ has no finite solution on $\lambda > 0$.

Final result and interpretation. For $T > 0$, both methods give $n/T = 1/\bar{X}$, and both give the estimate 1 for the observed sample. For $T = 0$, the likelihood is unbounded and the moment equation has no finite solution. The methods' agreement on positive-total samples is a feature of this model, not a general identity between them.

Marks (8). Correct likelihood, support and parameter space, 2; correct log-likelihood and score equation for $T > 0$, 1; correct casewise MLE conclusion, 1; valid maximum and boundary check, 1; numerical estimate, 1; method-of-moments derivation, 1; correct relationship interpretation, 1.

Common mistakes. Treating λ as the exponential mean rather than the rate; dropping the factor λ^n; using the score equation without checking the maximum or the $T = 0$ case; claiming method-of-moments and maximum-likelihood estimators always agree.

`W07-Q10-PA` **(M8 retrieval)**

Method and conditions. Use the leading-principal-minor test for a symmetric 2×2 matrix.

The leading principal minors are

$$\Delta_1 = -4 < 0$$

and

$$\Delta_2 = \det(\mathbf{A}) = (-4)(-2) - 2^2 = 4 > 0.$$

The alternating signs $\Delta_1 < 0$, $\Delta_2 > 0$ show that $\mathbf{A}$ is negative definite. Hence $q(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} < 0$ for every non-zero $\mathbf{x}$.

For

$$f(\mathbf{x}) = q(\mathbf{x}) + \mathbf{b}^{\mathsf{T}}\mathbf{x} + c,$$

the Hessian is $2\mathbf{A}$, which is also negative definite.

Final result. The quadratic form is negative definite, and f is strictly concave on $\mathbb{R}^2$.

Marks (3). Correct two principal minors, 1; correct definiteness classification, 1; correct Hessian and strict-concavity conclusion, 1.

Common mistakes. Using only the determinant; calling the form positive definite because the determinant is positive; forgetting the factor two in the Hessian.

`W07-Q10-PB` (`M9` retrieval)

Method and conditions. Solve the column-gradient condition and use the Hessian for a global sufficiency conclusion.

For

$$f(x, y) = 4x + 2y - x^2 - y^2,$$

the column gradient is

$$\nabla f(x, y) = \begin{bmatrix} 4 - 2x \\ 2 - 2y \end{bmatrix}.$$

Stationarity gives $x = 2$ and $y = 1$. The Hessian is

$$\mathbf{H}_f = -2\mathbf{I}_2,$$

which is negative definite on $\mathbb{R}^2$. The objective is strictly concave, so the stationary point is the unique global maximiser.

Final result. The unique global maximiser is $(2, 1)$.

Marks (3). Correct column gradient and stationarity condition, 1; correct candidate $(2, 1)$, 1; negative-definite Hessian, strict concavity and unique global classification, 1.

Common mistakes. Reporting a stationary point without classifying it; using a row gradient contrary to the course convention; claiming a global maximiser without a sufficiency argument.

`W07-Q10-PC` (`S6` retrieval)

Method and conditions. Use normal-sample closure and the variance of an iid sample mean, then standardise.

Because the observations are iid normal,

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{9}{36}\right) = \mathcal{N}\left(\mu, \frac{1}{4}\right).$$

The standard deviation of $\bar{X}$ is $1/2$. Therefore

$$\begin{aligned} \mathbb{P}(|\bar{X} - \mu| \leq 1) &= \mathbb{P}\left(\left|\frac{\bar{X} - \mu}{1/2}\right| \leq 2\right) \\ &= 2\Phi(2) - 1 \\ &= 2(0.9772) - 1 \\ &= 0.9544. \end{aligned}$$

Final result.

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{1}{4}\right), \qquad \mathbb{P}(|\bar{X} - \mu| \leq 1) = 0.9544.$$

Marks (2). Correct sampling distribution including variance, 1; correct standardisation and probability, 1.

Common mistakes. Dividing the standard deviation rather than the variance by 36; using variance $9/6$; calculating a one-sided probability.

Self-test solutions and error check

Solutions and error check

Solution to `W07-ST01-PA (M10)` Method. Use the stated equality encoding, solve the first-order system, then use strict concavity and the envelope theorem.

The Lagrangian is

$$\mathcal{L} = -(x-1)^2 - (y-2)^2 + \mu(a-x-y).$$

Stationarity gives

$$-2(x-1) - \mu = 0, \qquad -2(y-2) - \mu = 0.$$

Thus

$$x = 1 - \frac{\mu}{2}, \qquad y = 2 - \frac{\mu}{2}.$$

Feasibility gives $3 - \mu = a$, so

$$\mu^*(a) = 3 - a, \qquad x^*(a) = \frac{a-1}{2}, \qquad y^*(a) = \frac{a+1}{2}.$$

The objective Hessian is $-2\mathbf{I}_2$, so strict concavity and the affine constraint make this the unique global maximiser. The value function is

$$V(a) = -\frac{(a-3)^2}{2}.$$

Therefore

$$V'(4) = 3 - 4 = -1 = \mu^*(4) = \left.\frac{\partial \mathcal{L}}{\partial a}\right|_{a=4,*}.$$

Marks (5). Correct Lagrangian and stationarity, 1; correct optimisers, 1; correct multiplier, 1; sufficiency, 1; value and envelope check at $a = 4$, 1.

Common mistakes. Reversing the multiplier sign despite the stated encoding; solving only at $a = 4$; omitting the strict-concavity sufficiency condition.

Solution to `W07-ST02-PA (M11)` Method. Write both bounds as non-negative constraints and apply all four groups of Kuhn–Tucker conditions.

With $g_1(x) = x$ and $g_2(x) = 2 - x$,

$$\mathcal{L} = 6x - x^2 + \lambda_1 x + \lambda_2(2-x).$$

The conditions are

$$6 - 2x + \lambda_1 - \lambda_2 = 0,$$

$$x \geq 0, \qquad 2 - x \geq 0, \qquad \lambda_1, \lambda_2 \geq 0,$$

$$\lambda_1 x = 0, \qquad \lambda_2(2 - x) = 0.$$

The interior stationary point $x = 3$ violates the upper bound. At $x = 0$, stationarity would require a negative lower-bound multiplier. At $x = 2$, complementary slackness gives $\lambda_1 = 0$, and stationarity gives $\lambda_2 = 2$.

The objective is strictly concave on the convex feasible interval.

Final result.

$$x^* = 2, \qquad \lambda_1^* = 0, \qquad \lambda_2^* = 2,$$

and this is the unique global maximiser.

Marks (5). Correct constraints and Lagrangian, 1; complete conditions, 2; correct point and multipliers, 1; sufficiency, 1.

Common mistakes. Writing $x - 2 \geq 0$ for the upper bound; omitting the lower-bound multiplier; accepting the infeasible unconstrained point.

Solution to `W07-ST03-PA (S7)` Method. Use the sample-mean expectation and variance, then apply the bias–variance decomposition.

$$\mathbb{E}_\theta[\hat{\theta}_n] = \theta - \frac{1}{n},$$

so

$$\text{Bias}_\theta(\hat{\theta}_n) = -\frac{1}{n}.$$

The deterministic adjustment has zero variance, giving

$$\text{Var}_\theta(\hat{\theta}_n) = \frac{4}{n}.$$

Therefore

$$\text{MSE}_\theta(\hat{\theta}_n) = \frac{4}{n} + \frac{1}{n^2}.$$

The mean squared error tends to zero, which implies $\hat{\theta}_n \xrightarrow{p} \theta$. The estimator is consistent.

Marks (5). Bias, 1; variance, 1; mean squared error, 2; valid consistency conclusion, 1.

Common mistakes. Reporting positive bias; adding $1/n$ to the variance; forgetting to square the bias; claiming the estimator is unbiased.

Solution to `W07-ST04-PA (S7)` Method. Form the iid Poisson likelihood on $\lambda > 0$, use the log-likelihood score, and check curvature.

For observed counts $x_i \in \{0, 1, 2, ...\}$,

$$L(\lambda;\mathbf{x}) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\lambda^{\sum_i x_i}}{\prod_i x_i!}.$$

Ignoring terms constant in λ, the log-likelihood is

$$\ell(\lambda;\mathbf{x}) = -n\lambda + \left(\sum_i x_i\right)\log\lambda + \text{constant}.$$

For a positive total count, the score equation gives

$$-n + \frac{\sum_i x_i}{\lambda} = 0,$$

so

$$\hat{\lambda}_{\text{MLE}} = \bar{X}.$$

The second derivative is

$$-\frac{\sum_i x_i}{\lambda^2} < 0,$$

so this is the unique maximum. For the sample $(0, 1, 2, 1)$, $\bar{x} = 1$, and the estimate is 1.

Because $\mathbb{E}_\lambda[X_i] = \lambda$, the one-moment equation also gives

$$\hat{\lambda}_{\text{MOM}} = \bar{X}.$$

Marks (5). Correct likelihood or log-likelihood, 1; score equation, 1; MLE and maximum check, 1; numerical estimate, 1; method-of-moments estimator, 1.

Common mistakes. Treating the Poisson parameter as a variance unrelated to its mean; dropping the factor $n\lambda$; omitting the maximum check; using the sample total instead of the sample mean.

Record the first cause of each error under `M10`, `M11` or `S7`, then reattempt the affected part from a blank page in two or three days.